MD MAHFUZUR RAHMAN

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Research Summary

- Biomedical-focused Mechanical Engineer (Ph.D. candidate, ODU, expected March 2026) with 6+ years of academic research experience in developing, testing and validating affordable, wearable sensing systems for non-invasive cardiovascular health assessment — bridging the gap between laboratory-grade clinical instruments and accessible home-use devices. Conducting NSF-funded research (Grant #1936005) with 6 peer-reviewed publications (4 as first author).
- Experienced in end-to-end device prototyping — microfluidic tactile sensors and accelerometer-based handheld and wearable pulse acquisition systems — encompassing microfabrication, custom analog front-end circuitry (transimpedance amplifier, demodulation stage), and full sensor-to-DAQ system integration
- Expertise in cardiovascular biomechanical modeling and advanced signal-processing algorithms to study arterial pulse dynamics and motion artifacts, integrating FFT, EMD, CDM, HHT, and HVD for time-frequency analysis of nonstationary physiological signals
- Collaborated across academic and clinical settings on multiple IRB-approved human-subject studies – including NSF-funded projects, clinical research with Sentara Health Research Center, and joint research with the Department of Kinesiology and Health Sciences, University of Waterloo (Canada) – supporting protocol development, physiological data acquisition, and noninvasive cardiovascular health assessment using developed sensing systems

Education

Ph.D., Mechanical & Aerospace Engineering — Expected April 2026
Old Dominion University, Norfolk, VA | GPA: 3.87/4.0

- *Dissertation:* "Time-Frequency Analysis of Measured Arterial Pulse Signals for Motion Artifacts Removal"
- *Advisor:* Dr. Zhili Hao | NSF Grant #1936005

M.S., Applied Physics, Electronics & Communication Engineering — March 2012
University of Dhaka, Bangladesh | GPA: 3.73/4.0 (WES)

B.S., Applied Physics, Electronics & Communication Engineering — September 2010
University of Dhaka, Bangladesh | GPA: 3.81/4.0 (WES)

Publications

Journal Articles (Peer-Reviewed)

- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part I: Accelerometers and PPG Sensors. *Sensors*, 25(18), 5710. doi:10.3390/s25185710 [IF ≈ 3.5; SCIE; Q2]
- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part II: Tactile Sensors. *Sensors*, 25(18), 5700. doi:10.3390/s25185700 [IF ≈ 3.5; SCIE; Q2]
- **Rahman, M.M.**, et al. (2025). Motion Artifacts (MA) At-Rest in Measured Arterial Pulse Signals: Time-Varying Amplitude in Each Harmonic and Non-Flat Harmonic-MA-Coupled Baseline. *Biosensors*, 15(9), 578. doi:10.3390/bios15090578 [IF ≈ 5.6; SCIE; Q1]
- **Rahman, M.M.**, Hasan, M., May, J.F., Herre, J.M., Reynolds, L., & Hao, Z. (in press). Radial and carotid arterial pulse signals for assessing cardiovascular function at rest and during post-exercise recovery in a heart transplant patient: a case study. *Medical Sensors & Imaging* (OP Publishing) [Accepted; doi:10.1088/2977-8425/ae4b40; New Journal]
- Toraskar, S., **Rahman, M.M.**, & Hao, Z. (2026). Pulse Measurement Alters the True Pulse Signal in an Artery:

A Coupled String-SDOF Model. *ASME J. Engineering and Science in Medical Diagnostics and Therapy*, 9(1), 011207. doi:10.1115/1.4069099 [IF ≈ 1.1; ESCI; Q3]

- Hao, Z., **Rahman, M.M.**, et al. (2023). Axial Wall Displacement at the Common Carotid Artery is Associated With the Lamb Waves. *ASME J. Engineering and Science in Medical Diagnostics and Therapy*, 6(1), 011008. doi:10.1115/1.4056267 [IF ≈ 1.1; ESCI; Q3]

Conference Proceedings

- **Rahman, M.M.**, et al. (2021). Measurement of Post-Exercise Response of Local Arterial Parameters Using an Adjustable Microfluidic Tactile Sensor. *IEEE EMBC 2021*, pp. 1284–1287. [Virtual; Presented]. doi:10.1109/EMBC46164.2021.9630432
- **Rahman, M.M.**, et al. (2020). Improved Vibration-Model-Based Analysis for Estimation of Arterial Parameters From Noninvasively Measured Arterial Pulse Signals. *ASME IMECE 2020*, V005T05A079. [Virtual; Presented]. doi:10.1115/IMECE2020-24551

Research Experience

Graduate Research Assistant — NSF Grant #1936005 Sept 2019 – Present
Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Project 1: Soft Polymer Microfluidic Tactile Sensor — Fabrication and Characterization

- Designed and fabricated soft polymer (PDMS) based microfluidic tactile sensors end-to-end using a standard polymer-based process: CAD design of electrode patterns and micro-channel (AutoCAD); deposited and patterned 100 nm/10 nm Au/Cr electrodes on Pyrex slides via photolithography (spin coating, UV exposure, development), sputter deposition (EMS300T D), and lift-off; formed SU-8 micro-channel molds (100–150 μm channel height); cast PDMS microstructures using a 10:1 elastomer-to-curing-agent ratio (Sylgard 184); bonded PDMS-to-Pyrex via oxygen plasma treatment (PDC-001); and filled microchannels with electrolyte via syringe through reservoir holes
- 3D-printed microchannel molds being tried as a cost- and time-effective alternative to SU-8-based molds, reducing fabrication cycle time by eliminating spin-coating, UV exposure, development, and baking steps while maintaining microchannel geometry fidelity
- Designed five single-channel PCBs (KiCad) for analog readout circuit: transimpedance amplifier (OPA656U), AD835-based demodulation stage(multiplier and 3rd-order 100 Hz low-pass filter; five parallel channels enable simultaneous readout of all transducers; NI DAQ digitization with real-time LabVIEW monitoring
- Characterized sensor static performance using a micropositioner-controlled circular probe (4 mm diameter, 0.2 μm displacement resolution): mapped distributed load detection capability, resistance-change-to-load sensitivity, spatial resolution (1.5 mm transducer spacing)
- Extended sensor platform to arterial pulse signal acquisition: characterized dynamic frequency response, sensitivity, linearity, and repeatability for pulse waveform fidelity; results validated in *Sensors* 2025 (Parts I & II)

Project 2: Noninvasive Physiological Measurement — Arterial Stiffness & Motion Artifacts

- Conducted IRB-approved studies on healthy participants (IRB24-166) and cardiac rehabilitation patients (IRB Approval #19-04-FB-0100) using microfluidic tactile sensors at the radial and carotid arteries
- Developed and validated an analytical model (tissue-contact-sensor 1DOF system) showing that motion artifacts generate both additive baseline drift *and* time-varying system parameters—challenging the adequacy of current wavelet/EMD/CSE removal algorithms (*Biosensors* 2025; *Sensors* 2025)
- Implemented MATLAB pipelines using FFT, Hilbert-Huang Transform (HHT), Empirical Mode Decomposition (EMD), Complex Demodulation (CDM), Hilbert Vibration Decomposition (HVD), and ODE45 for validation
- Extracted and validated arterial parameters (elasticity, viscosity, radius, PWV) from single-sensor pulse signals using vibration-model-based analysis (IEEE EMBC 2021; ASME IMECE 2020)

Project 3: Comprehensive Arterial Health Assessment Across BMI Groups (IRB #1452174-27) — Ongoing

- Designed and executed a multi-visit IRB-approved clinical study measuring arterial pulse signals in healthy male subjects across varying BMI ranges (<25 and 30–40 kg/m^2) using two microfluidic tactile sensor designs at the radial and carotid arteries

- Acquired multimodal cardiovascular data at rest and across six post-exercise time points (immediately, 10, 15, 20, 30, and 60 min post-exercise) using the microfluidic tactile sensor, Doppler ultrasound (arterial diameter, velocity, blood flow), applanation tonometry (vascular stiffness via ECG-gated tonometer), and brachial blood pressure measurement
- Evaluated reproducibility of two sensor designs (differing in PDMS thickness) for pulse waveform acquisition quality and derived physiological parameters including heart rate, respiration rate, augmentation index, and pulse wave velocity
- Investigated correlations between tactile sensor-derived arterial indices and established clinical reference measurements (Doppler ultrasound, tonometry, blood pressure) to validate sensor performance in a structured clinical setting
- Study temporarily paused; protocol and dataset infrastructure fully established for resumption

Project 4: Feasibility Study — Accelerometer-Based Carotid Arterial Pulse Acquisition (IRBnet #1744951-11)

- Designed a custom flexible PCB integrating a single ST Microelectronics analog accelerometer (LIS344ALH) and subsequently a digital accelerometer (LIS2DW12) for 3-axis carotid arterial wall acceleration capture; raw arterial pulse signal amplitude at the carotid artery is in the range of 20–30 $\mathrm{mV_{PP}}$ with an approximately 2 V DC offset, posing significant signal conditioning challenges
- Developed analog front-end (AFE) circuitry using op-amp-based signal conditioning to extract the true arterial pulse signal (APG) from the raw accelerometer output — removing DC offset without introducing phase delay — followed by amplification and NI DAQ digitization with real-time LabVIEW monitoring and MATLAB post-processing for physiological parameter extraction (heart rate, PWV)
- Conducted an IRB-approved single-visit feasibility study to evaluate whether analog and digital accelerometers can reliably capture a clear carotid arterial pulse signal prior to committing to a full-scale performance study
- Measured carotid pulse signals at two anatomical sites per side of the neck (proximal and distal to the head) on both left and right sides sequentially — 3 repeated 1-min recordings per site per accelerometer — enabling inter-site transit time estimation for local PWV calculation
- Compared analog (LIS344ALH) vs. digital (LIS2DW12) accelerometer performance at identical measurement sites under medical-tape fixation, with simultaneous brachial blood pressure and heart rate reference measurements (Mindray instrument)
- Identified fundamental limitations of accelerometer-based carotid pulse sensing — tape-fixation motion coupling, low signal-to-noise ratio at the neck surface, and DC offset instability under varying contact pressure — informing the analytical motion artifact framework subsequently published in *Sensors* 2025 Part I

Technical Skills

- **Microfabrication:** Photolithography (spin coating, UV exposure, development); Au/Cr thin-film deposition (sputtering); lift-off; plasma treatment; soft lithography (SU-8 based micro-channel mold fabrication)
- **Sensor Design & Electronics:** AutoCAD (electrodes and microchannel layout); Autodesk Inventor (3D mold design for microchannel); Multisim (circuit simulation); KiCad (PCB design for analog front-end signal conditioning circuitry like: transimpedance amplifier, multiplier, low-pass filter, adder, subtractor) ; LabVIEW-based real-time DAQ and monitoring
- **Signal Processing & Computation:** MATLAB (Filter Design, Smoothing, Cubic Spline Interpolation, Continuous Wavelet Transforms (CWT), Discrete Wavelet TransformDWT, Wavelet Decomposition (WAVEDEC), FFT, ODE45, HHT, EMD, CDM, HVD, custom arterial pulse analysis pipelines), Python, Simulink, LabVIEW (Acquire, visualize and record data), SPSS (T-tests, ANOVA)
- **Clinical & Human Subjects Research:** IRB protocol design and compliance; REDCap (data collection, entry, and quality management); pulse signal acquisition at radial and carotid arteries; cardiac rehabilitation patient studies (Sentara Health Research Center);blood pressure, heart rate, and SpO_2 measurement (Mindray); applanation tonometry; doppler ultrasound
- **Cross-Functional Collaboration & Translational Research:** Partnered with clinical and academic collaborators at Sentara Health Research Center and the University of Waterloo (Canada) on NSF-funded, IRB-

approved human subject studies; contributed to patient recruitment and screening, protocol design, physiological data acquisition, and experimental documentation supporting noninvasive cardiovascular health assessment

Teaching Experience

Graduate Teaching Assistant — Mechanical & Aerospace Engineering Sept 2019 – Present
Old Dominion University, Norfolk, VA

- Delivered laboratory instruction, graded assignments, and mentored undergraduate students across core mechanical engineering courses including dynamics, mechanics of materials, and instrumentation

Lecturer(Full Time) — Electronics & Telecommunication Engineering Oct 2014 – Mar 2017
Atish Dipankar University of Science & Technology, Dhaka, Bangladesh

- Taught undergraduate courses in electronics, telecommunications, and microprocessors

Lecturer(Part-Time) — Computer Science and Engineering (CSE) Aug 2016 – Mar 2017
Daffodil Institute of Information Technology, Dhaka, Bangladesh

- Instructed electrical engineering and microprocessor courses

Grants & Funding

- NSF Grant #1936005 — Noninvasive Cardiovascular Sensing Systems | PI: Dr. Zhili Hao, ODU | Role: Graduate Research Assistant | 2019–Present

Honors & Awards

- Tiwari Endowed Graduate Scholarship, Old Dominion University 2021
- National Science & ICT (NSICT) Fellowship, Bangladesh 2010
- Merit-Based Scholarship, University of Dhaka 2008

Certifications

- Biomedical Research — Basic, CITI Program
  Issued March 2026 · Expires March 2029 · Credential ID: 72079387 *(Active)*
- Biomedical Research — Refresher 1, CITI Program
  Issued November 2021 · Expired November 2023 · Credential ID: 43904237
- Biomedical Research — Refresher 2, CITI Program
  Issued December 2023 · Expired December 2025 · Credential ID: 57666724
- Biomedical Responsible Conduct of Research, CITI Program
  Issued October 2020 · Credential ID: 39136829

Professional & Community Service

- Volunteer, Virginia Medical Reserve Corps — COVID-19 vaccination clinics, health screenings, patient intake May 2020–Present
- Author, IEEE Engineering in Medicine & Biology Conference (EMBC 2021)
- Published in ASME Journal of Engineering and Science in Medical Diagnostics and Therapy

Industry Experience

Manager — Merchandising & Production Apr 2017 – Aug 2019

Afiya Knitwear Ltd., Dhaka, Bangladesh

- Managed end-to-end production planning, costing, quality compliance, and vendor coordination for international apparel manufacturing contracts
- Developed cross-functional project management and operational efficiency skills in multi-stakeholder, deadline-driven environments

References

Available upon request.
Primary academic reference: Dr. Zhili Hao, Professor, Dept. of Mechanical & Aerospace Engineering, Old Dominion University (zlhao@odu.edu)